

DATA ANALYST

Intenship project 2

DESCRIPTION

This task involves analyzing an e-commerce dataset to identify return patterns and high-risk products. The objective is to calculate return rates, classify products based on risk levels, and build an interactive Power BI dashboard to visualize trends by category, region, and product, enabling data-driven business insights and decision-making.

PREPARED BY

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MY WORK

In this project, I performed data analysis on an e-commerce dataset to evaluate product return behavior. I first cleaned and prepared the dataset, then calculated key metrics such as total orders, total returns, and return rate. I created a risk classification system to identify high-risk products based on return patterns. Using Power BI, I developed an interactive dashboard with KPI cards, category-wise and region-wise return analysis, trend visualization, and a high-risk product table. I also added slicers for dynamic filtering. The final dashboard provides clear insights to help businesses reduce return rates and improve operational decisions.

DATASET USED

MY GOOGLE COLAB WORK TO CLEAN RETURN AND ORDER DATASET

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("ecommerce_data.csv")
df.head()
```

	customer_id	customer_name	city	region	order_id	order_date	product_id	product_name
0	1	Aarav Sharma	Delhi	North	1001	1/1/2024	P01	Wireless Mouse
1	2	Diya Patel	Ahmedabad	West	1002	1/2/2024	P02	Bluetooth Headphones
2	3	Rohan Verma	Mumbai	West	1003	1/3/2024	P03	Notebook
3	4	Ananya Iyer	Chennai	South	1004	1/4/2024	P04	Water Bottle
4	5	Kabir Singh	Delhi	North	1005	1/5/2024	P05	Office Chair

```
import numpy as np

df['Returned'] = np.random.choice([0,1], size=len(df), p=[0.8,0.2])
```

```
print(df.columns)
```

```
Index(['customer_id', 'customer_name', 'city', 'region', 'order_id',
       'order_date', 'product_id', 'product_name', 'category', 'quantity',
       'price', 'total_sales', 'Returned'],
      dtype='object')
```

```
df['Returned'].value_counts()
```

	count
Returned	
0	161
1	39

dtype: int64

```
print("Return Rate:", df['Returned'].mean())
```

Return Rate: 0.195

```
df.groupby('category')['Returned'].mean().sort_values(ascending=False)
```

Returned	
category	
Stationery	0.275
Accessories	0.250
Electronics	0.200
Home	0.150
Furniture	0.100

dtype: float64

```
df.groupby('region')['Returned'].mean().sort_values(ascending=False)
```

Returned	
region	
East	0.30
West	0.25
North	0.15
South	0.15

dtype: float64

```
df.groupby('product_name')['Returned'].mean().sort_values(ascending=False)
```

Returned	
product_name	
Pen Set	0.30
Backpack	0.25
Bluetooth Headphones	0.25
Notebook	0.25
Smart Watch	0.25
Water Bottle	0.20
Coffee Mug	0.15
Desk Lamp	0.10
Office Chair	0.10
Wireless Mouse	0.10

dtype: float64

```
df['Order_Value'] = df['quantity'] * df['price']
```

```
df_model = df.drop(['customer_name', 'order_date', 'product_name'], axis=1)
```

```
df_model = pd.get_dummies(df_model, drop_first=True)
```

```
X = df_model.drop('Returned', axis=1)
y = df_model['Returned']
```

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
from sklearn.linear_model import LogisticRegression
```

```
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.
```

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
```

LogisticRegression

```
LogisticRegression(max_iter=1000)
```

```
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
y_pred = model.predict(X_test)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
print(confusion_matrix(y_test, y_pred))
```

```
Accuracy: 0.9
```

```
[[36 0]
```

```
 [ 4 0]]
```

```
df_model['Return_Probability'] = model.predict_proba(X)[:,-1]
```

```
high_risk = df_model[df_model['Return_Probability'] > 0.7]
```

```
high_risk.to_csv("high_risk_products.csv", index=False)
```

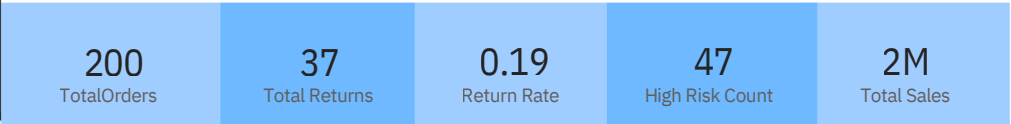
DATASET WITH RETURN AND HIGH RISK CSV FROM ANALYSIS USING PYTHON ON GOOGLE COLAB

- High_risk
- E-commerce dataset with returns

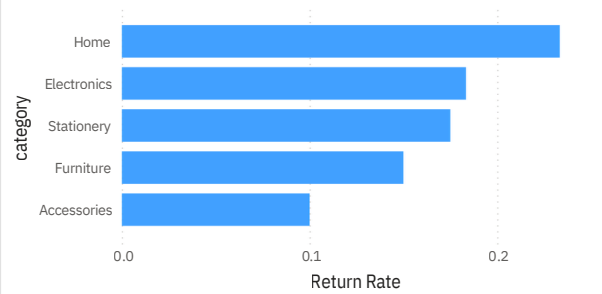
POWER BI DASHBOARD



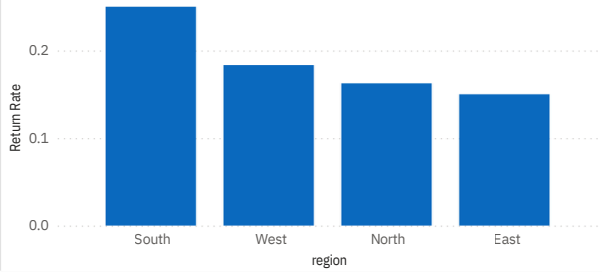
E_Commerce_Return_Rate_Analysis



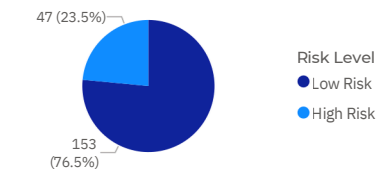
CATEGORY ANALYSIS



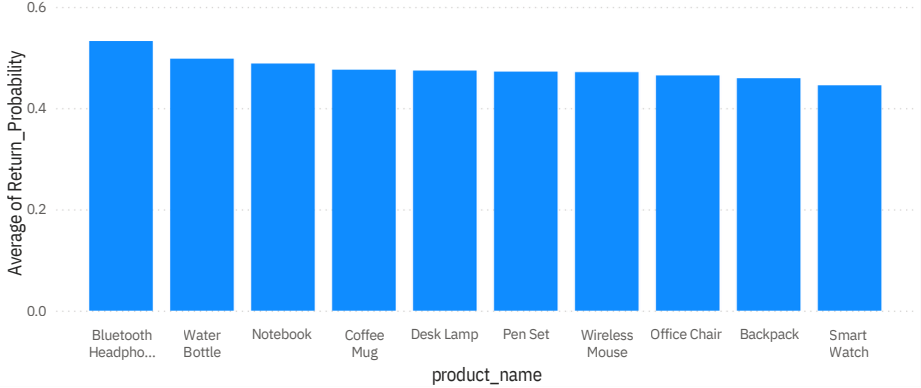
REGION ANALYSIS



RISK DISTRIBUTION



PRODUCT ANALYSIS



category	region	TotalSales	Sum of Return_Probability	product_name
Accessories	West	99960	9.18	Backpack
Electronics	West	79960	10.65	Bluetooth Headphones
Home	North	34900	9.52	Coffee Mug
Home	South	17980	9.49	Desk Lamp
Stationery	West	3600	9.76	Notebook
Furniture	North	749900	9.29	Office Chair
Stationery	East	7200	9.45	Pen Set
Electronics	North	479920	8.90	Smart Watch
Home	South	23920	9.96	Water Bottle
Electronics	North	15980	9.43	Wireless Mouse
Total		1513320	95.63	